

ABHIJAY KUMAR SHAH

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Education

Vellore Institute of Technology B.Tech in Computer Science & Engineering	Jul 2026 (Expected) CGPA: 8.44 / 10.0
DAV Public School, Dudhichua Senior Secondary (Class XII)	2019–2021 CGPA: 9.20 / 10.0
DAV Public School, Surya Vihar Secondary (Class X)	2018–2019 CGPA: 9.60 / 10.0

Technical Skills

Immersive Tech & Graphics: Unity3D (C# Scripting), Unreal Engine (Core Logic), 3D Content Pipelines, WebGL/Three.js
Core Engineering: Advanced OOP (C++, Java, Python), Data Structures (DSA), Design Patterns, SOLID Principles
Visualization & AI/ML: AI-enabled Visualization (Gemini/GPT-4), Computer Vision Basics, Python AI Data Pipelines
Software Engineering: Agile/Scrum (Daily Stand-ups), Technical Documentation, Unit Testing (Jest), API Testing (Postman)
Systems & Integration: Blender (3D Asset Integration), RESTful APIs, WebSockets (Real-time Sync), Git/GitHub, Docker

Projects

MultiSync — High-Performance Visualization Engine | *Vanilla JS, OOP, Graphics Optimization* [\[Link\]](#)

- Engineered a real-time visualization engine synchronizing **9 concurrent HD video streams** at 10x playback speed, demonstrating mastery over high-bandwidth graphics processing and state management.
- Applied strict **Object-Oriented Programming (OOP) encapsulation** to manage independent player instances, mirroring the component-based architecture required for Unity3D/XR pipelines.
- Implemented hardware-specific **performance optimization** and memory leak prevention for high-load environments, ensuring zero frame drops during multi-stream synchronization.

ProjectHub — AI-Powered Enterprise Visualization | *React, TypeScript, Node.js, OOP* [\[Link\]](#)

- Developed a production-grade system using **Object-Oriented Design principles** to manage role-based (Student/Faculty) document workflows and immersive visual feedback loops.
- Architected a secure **REST API with 15+ endpoints**, integrating Cloudinary and MongoDB Atlas to manage complex 3D asset data and enterprise-level document persistence.
- Systematically **debugged critical production issues**, including network protocol (IPv4/IPv6) conflicts and header misconfigurations, ensuring 100% system availability in a cloud environment.
- Integrated an **AI-enabled visualization engine** using structured LLM prompts to validate academic document compliance against industry-standard formatting benchmarks.

Rately — Multi-Tenant Simulation & Analytics Platform | *Node.js, MongoDB, RBAC* [\[Link\]](#)

- Designed a scalable **Multi-Tenant Architecture** with Role-Based Access Control (RBAC), implementing inheritance patterns to isolate sensitive enterprise data across different user hierarchies.
- Optimized backend query performance using **Data Structures (Indexing/Aggregation)** to handle real-time store rating visualizations for high-concurrency environments.
- Ensured **System Reliability** by implementing rate limiting and security hardening (Helmet.js), following enterprise software engineering best practices for mission-critical applications.

Experience / Leadership – Achievements

CatCatchCode **catcatchcode.online – Jul 2025 – Present**

Co-Founder & Technical Lead — VIT

[\[Link\]](#)

- Founded an ed-tech initiative providing placement guidance, curated study material, and senior mentorship for students.
- Led a 2-member team to produce 100+ technical learning videos, impacting 100+ students.
- Designed beginner-to-intermediate learning roadmaps for DSA, Full-Stack development, and AI automation.
- Expanded content across LinkedIn, YouTube, Instagram, and GitHub, building an early-stage learning community.

Satsang Vachan Satsang Seva **satsangvachansatsangseva.online – Sept 2025 – Present**

Co-Founder & Project Lead – Digital Preservation & Social Impact Initiative

[\[Link\]](#)

- Leading a digital preservation initiative to convert decades of archival literature into modern visual and multimedia formats.
- Spearheading a digitization initiative to preserve 50+ years of archival literature for a global community of 200 Million+ followers.
- Collaborating with contributors to produce posters, magazines, podcasts, and animated content for large-scale audiences.
- Applying AI-assisted summarization to transform long-form textual material into concise, accessible visuals.
- Designing a scalable content platform focused on accessibility, maintainability, and positive social impact.

Certifications

Python Essentials (VIT) — May 2024 [\[Link\]](#)

MERN Full Stack Certificate — Apr 2025 [\[Link\]](#)